

Agile Backlog Creation & Sprint Simulation in Jira

Backlog management, story point estimation,
sprint simulation and burndown analysis

PROBLEM STATEMENT #23

Hyperlocal Courier Dispatch & Tracking Engine

NAME

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SECTION

5th Semester B

DATE OF COMPLETION

26 August 2026

JIRA PROJECT

Hyperlocal Courier Dispatch

PROJECT KEY

HCD

Lab 2 Deliverable

Agile Backlog Management, Story Point Estimation, Sprint Simulation and Burndown Analysis

Project Overview

The **Hyperlocal Courier Dispatch & Tracking Engine** is an on-demand courier platform designed to manage local parcel delivery from request creation through final delivery verification.

The system allows sender clients to create courier requests, assigns nearby delivery riders, supports delivery tracking and live rider location monitoring, optimizes routes for multiple delivery stops, and verifies successful delivery using destination-based OTP verification.

For Lab 2, the functional requirements identified during Lab 1 were converted into Agile Epics and User Stories in Jira. The backlog was prioritized, estimated using Fibonacci-based Story Points, organized into two one-week sprints, and progressed through the Jira Scrum board.

Both planned sprints were completed successfully.

JIRA PROJECT DETAILS

PROJECT Hyperlocal Courier Dispatch	PROJECT KEY HCD	BOARD HCD board
METHOD Scrum	SPRINTS COMPLETED 2	TOTAL USER STORIES 8
TOTAL STORY POINTS 31	EPICS 3	WORKFLOW To Do → In Progress → Done

JIRA PROJECT / WORKSPACE LINK

Jira Project Link
<https://pes1ug24cs122.atlassian.net/jira/software/c/projects/HCD/summary>
The Jira workspace is live and available for instructor demonstration.

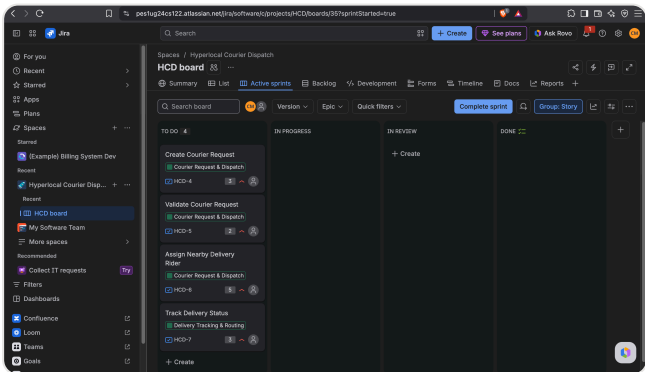
Epics and User Stories

EPICS

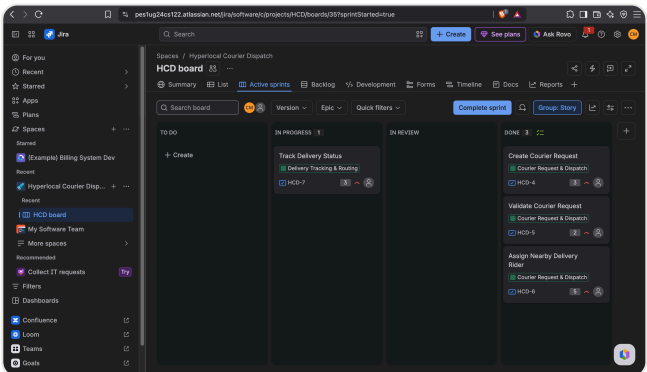
EPIC ID	EPIC
HCD-1	Courier Request & Dispatch
HCD-2	Delivery Tracking & Routing
HCD-3	Delivery Completion & Verification

USER STORIES

ID	USER STORY	EPIC
HCD-4	Create Courier Request	HCD-1
HCD-5	Validate Courier Request	HCD-1
HCD-6	Assign Nearby Delivery Rider	HCD-1
HCD-7	Track Delivery Status	HCD-2
HCD-8	View Live Rider Location	HCD-2
HCD-9	Optimize Delivery Route	HCD-2
HCD-10	Complete Delivery	HCD-3
HCD-11	Verify Delivery with OTP	HCD-3



Evidence A — HCD-4 to HCD-7 with parent Epic labels **Courier Request & Dispatch (HCD-1)** and **Delivery Tracking & Routing (HCD-2)**.



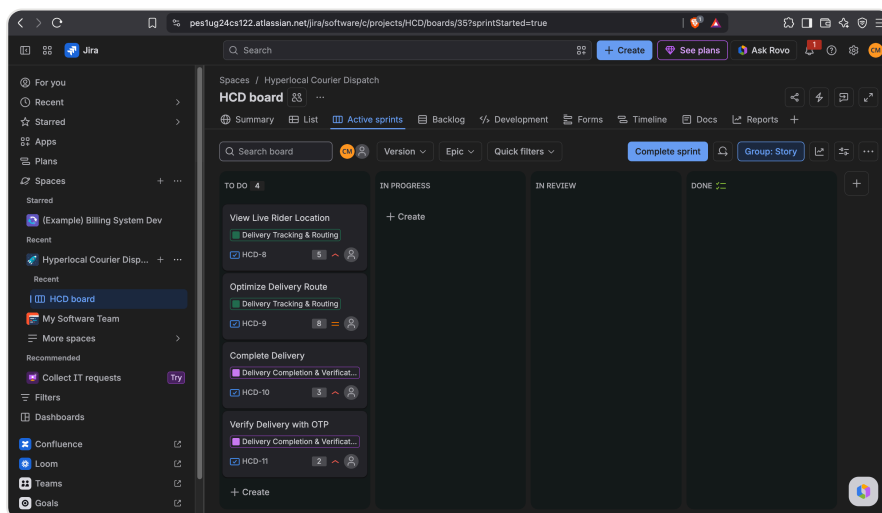
Evidence B — HCD-8 to HCD-11 with parent Epic labels **Delivery Tracking & Routing (HCD-2)** and **Delivery Completion & Verification (HCD-3)**.

Together, the two views cover all three Epics (HCD-1, HCD-2, HCD-3) and all eight User Stories (HCD-4 to HCD-11).

Story Point Estimation and Priorities

ID	USER STORY	PRIORITY	POINTS
HCD-4	Create Courier Request	High	3
HCD-5	Validate Courier Request	High	2
HCD-6	Assign Nearby Delivery Rider	High	5
HCD-7	Track Delivery Status	High	3
HCD-8	View Live Rider Location	High	5
HCD-9	Optimize Delivery Route	Medium	8
HCD-10	Complete Delivery	High	3
HCD-11	Verify Delivery with OTP	High	2
Total Story Points			31

Story Points were assigned using the Fibonacci-style scale (2, 3, 5, 8) according to the relative effort, complexity, uncertainty and risk associated with each User Story. Higher values were assigned to stories involving greater technical complexity, such as route optimization.



Evidence: Jira cards displaying the assigned Story Point values (5, 8, 3, 2) on the sprint board.

Sprint 1 — HCD Sprint 1

Sprint Name: HCD Sprint 1 | **Duration:** 1 Week

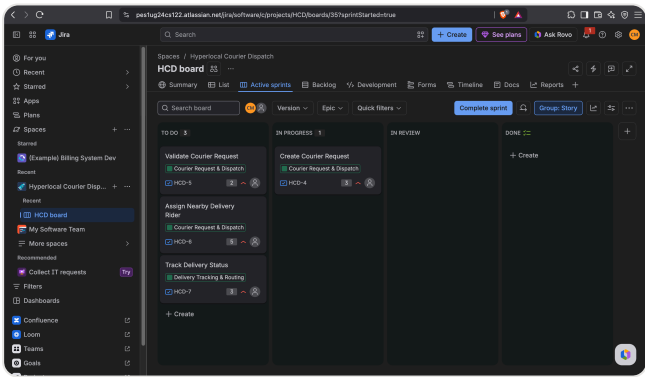
Sprint Goal: Complete the core courier request, validation, rider assignment, and delivery tracking flow.

SPRINT SCOPE

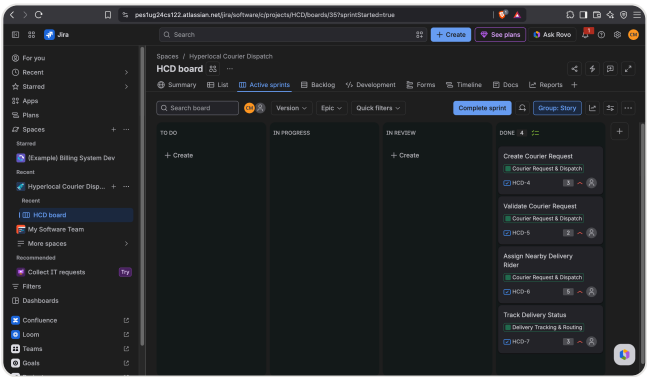
ID	USER STORY	POINTS
HCD-4	Create Courier Request	3
HCD-5	Validate Courier Request	2
HCD-6	Assign Nearby Delivery Rider	5
HCD-7	Track Delivery Status	3
Sprint Total		13

SPRINT PROGRESS

All four selected User Stories progressed through **To Do** → **In Progress** → **Done**. Sprint 1 was completed successfully with all 4 User Stories completed and all 13 committed Story Points delivered.



Screenshot A — Sprint 1 active board with stories moving from To Do into In Progress.



Screenshot B — Sprint 1 board with HCD-4 to HCD-7 completed in the Done column.

Sprint 2 — HCD Sprint 2

Sprint Name: HCD Sprint 2 | **Duration:** 1 Week

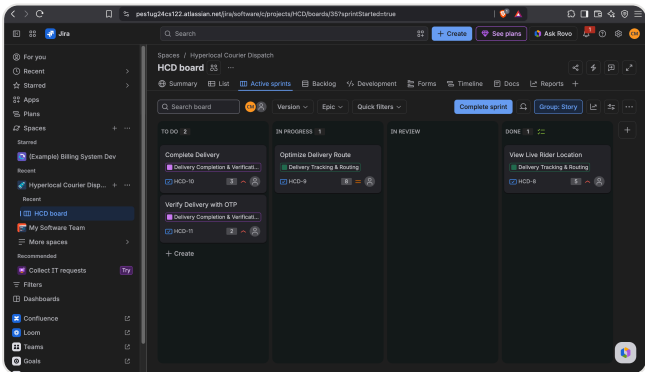
Sprint Goal: Complete delivery tracking, route optimization, and secure delivery verification through successful delivery completion.

SPRINT SCOPE

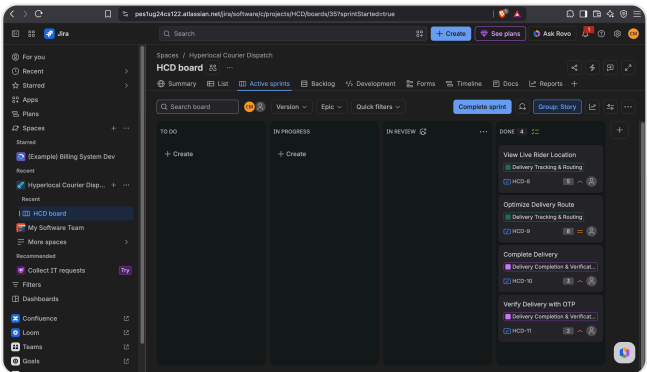
ID	USER STORY	POINTS
HCD-8	View Live Rider Location	5
HCD-9	Optimize Delivery Route	8
HCD-10	Complete Delivery	3
HCD-11	Verify Delivery with OTP	2
Sprint Total		18

SPRINT PROGRESS

All four selected User Stories progressed through **To Do** → **In Progress** → **Done**. Sprint 2 was completed successfully with all 4 User Stories completed and all 18 committed Story Points delivered.



Screenshot A — Sprint 2 active board with work distributed between To Do and In Progress.

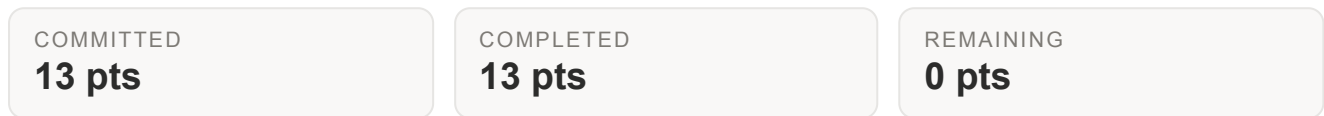


Screenshot B — Final Sprint 2 board with HCD-8 to HCD-11 all in the Done column.

Burndown Chart — Sprint 1

The Jira Burndown Chart was used to monitor the amount of remaining Story Points during each sprint. A burndown chart represents completed work against remaining work and helps assess whether the planned work is likely to be finished within the available sprint time.

Sprint 1 started with **13 Story Points** and reached **0 remaining Story Points** after all four User Stories (HCD-4 to HCD-7) were completed.



Evidence: Jira Burndown Chart for **HCD Sprint 1**, measured in Story Points, with the sprint goal shown above the chart.

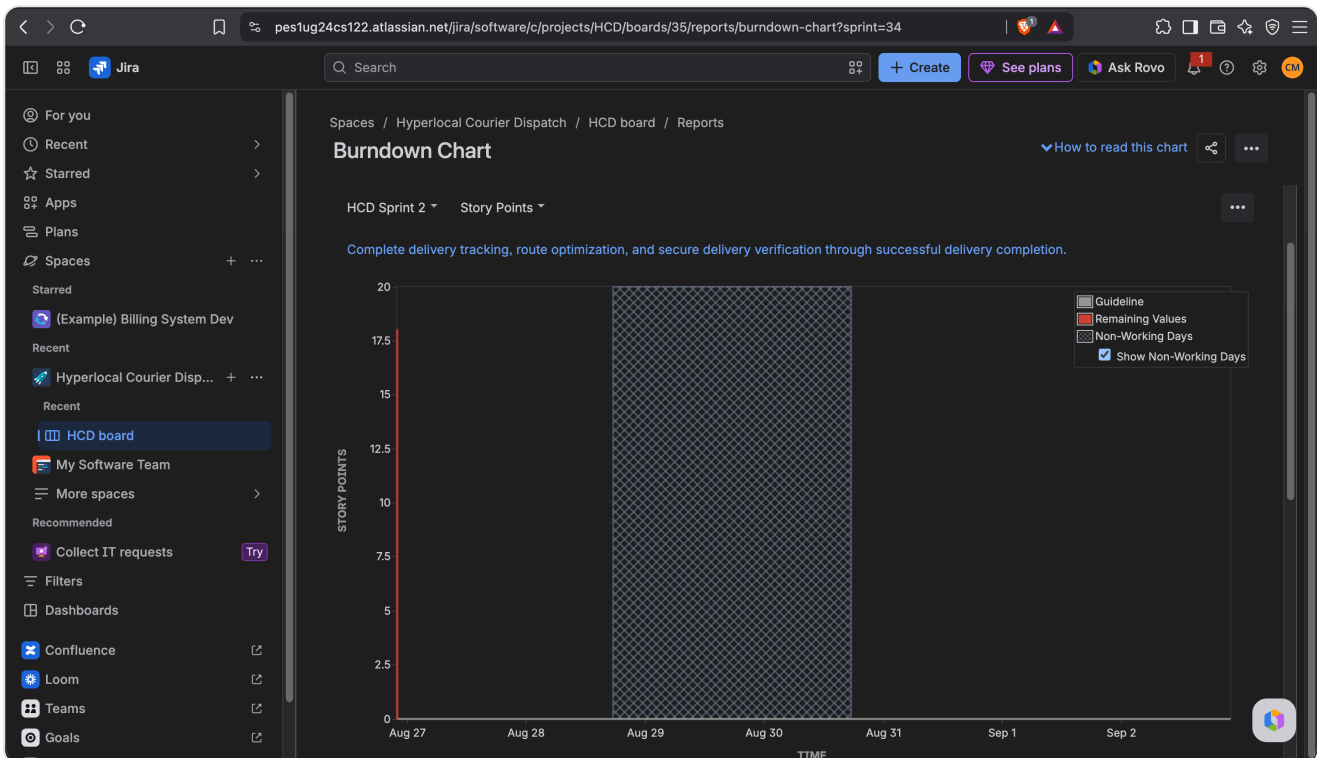
Burndown Chart — Sprint 2

Sprint 2 started with **18 Story Points** and reached **0 remaining Story Points** after all four User Stories (HCD-8 to HCD-11) were completed.

COMMITTED
18 pts

COMPLETED
18 pts

REMAINING
0 pts



Evidence: Jira Burndown Chart for **HCD Sprint 2**, measured in Story Points.

Observation (Both Sprints)

Both sprints reached completion with 0 remaining Story Points — 13 points in Sprint 1 and 18 points in Sprint 2. The actual work was completed rapidly because the sprints were simulated in a compressed session rather than being executed across the full one-week period. Therefore, both charts show a sharp reduction in remaining work rather than a gradual daily burndown.

Reflection

1 Did your estimations reflect the actual effort?

The estimates provided a reasonable relative representation of the effort required for the User Stories. Smaller stories such as OTP verification were assigned fewer Story Points, while the more complex route optimization story received a higher estimate of 8 points. Since the exercise was a simulated sprint, the estimates cannot be compared against real development hours, but they were useful for prioritizing and planning the sprint work.

2 Was your backlog well-prioritized?

Yes. The backlog was prioritized according to the importance of the functionality to the courier workflow. Core activities such as creating and validating courier requests, assigning riders, and tracking deliveries were given High priority, while route optimization was assigned Medium priority. This allowed the essential courier workflow to be completed first in Sprint 1.

3 How did your simulated sprint align with your plan?

The simulated sprints aligned successfully with the planned scope. Sprint 1 contained four stories totaling 13 Story Points, and all four were completed. Sprint 2 contained four stories totaling 18 Story Points, and all four were also completed. Both sprints followed the planned To Do → In Progress → Done workflow and were completed successfully.

4 What insights did the burndown chart give about your team's capacity?

The burndown charts for both sprints showed that the complete sprint scope could be finished successfully within the planned work. However, because the exercise was performed as a compressed simulation, the work was completed much faster than it would normally be during a real one-week sprint. The sharp burndown therefore demonstrates successful completion of the committed scope, but it should not be interpreted as a realistic measure of daily team velocity.

Lab 2 Summary

Lab 2 converted the functional requirements from Lab 1 into an Agile Jira backlog consisting of **three Epics** and **eight User Stories**. The User Stories were prioritized and estimated using Fibonacci-based Story Points for a total of **31 points**.

SPRINT OUTCOME

SPRINT	STORIES	POINTS	STATUS
HCD Sprint 1	4	13	Completed
HCD Sprint 2	4	18	Completed
Total	8	31	Completed

FINAL OUTCOME

The exercise demonstrated the use of Agile backlog management, Epic and User Story organization, Story Point estimation, Sprint planning, Scrum board workflow, and Burndown Chart analysis using Jira.

Jira Project / Workspace

<https://pes1ug24cs122.atlassian.net/jira/software/c/projects/HCD/summary>